SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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CSA37- SOFTWARE TESTING

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Requirement Analysis and Negative Test Scenarios and Test Cases for an Online Examination System

Aim

To study the requirements of an Online Examination System, and to prepare negative test scenarios and test cases that confirm the application handles invalid inputs and wrong user actions safely.

Problem Statement

An online examination system lets an examiner set up exams and lets candidates write them on a computer, with automatic marking. Many candidates use it at the same time, and the results affect their careers, so a defect can cause wrong marks, lost answers or unfair exams.

Testing is needed to check that the application:

- Candidates can log in, write an exam and get correct results.
- Invalid inputs, cheating attempts and unauthorized access are handled safely.
- Timing, submission and evaluation rules follow the requirements.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	Candidates log in with a register number and password. Examiners log in through a separate admin login.
FR2	The admin can create an exam with questions, marks, duration and a scheduled start and end time.
FR3	A candidate can start the exam only within the scheduled time window.
FR4	A countdown timer shows the remaining time, and the exam is submitted automatically when time ends.
FR5	Questions are shown one per page. The candidate can move to the next or previous question and mark questions for review.
FR6	Selected answers are saved automatically, and a candidate can change an answer before submitting.
FR7	The exam can be submitted only once. A confirmation is asked before the final submission.
FR8	Evaluation is automatic. One mark is given for each correct answer and 0.25 mark is deducted for each wrong answer.
FR9	After 3 warnings for switching tabs or windows, the exam is submitted automatically.
FR10	If the network breaks, the candidate can log in again and continue with the remaining time and saved answers.

ID	Requirement
FR11	Results are shown to candidates only after the admin publishes them.
FR12	The admin can view the result report. A candidate cannot open admin pages, and one account cannot be logged in on two devices at once.

2 Software and environment requirements

Item	Details
Application	Web application (Node.js / Java back end, React front end)
Database	MySQL (users, exams, questions, answers, results)
Operating system	Windows 10/11 or Linux
Browsers	Chrome, Firefox, Edge (latest versions)
Testing tools	Test case document (Excel), Selenium, JMeter, bug tracker (Jira)
Test data	Question bank of 20 MCQs, sample candidates and admin accounts, an exam scheduled for a fixed time window

Negative Test Scenarios

A negative test scenario checks that the system rejects invalid input or a wrong action, shows a clear message and stays stable and secure.

Scenario ID	Module	Test Scenario	Requirement
OE-NS01	Login	Log in with a wrong password	FR1
OE-NS02	Login	Submit blank credentials	FR1
OE-NS03	Start exam	Start the exam before the scheduled time	FR3
OE-NS04	Start exam	Start the exam after the window has closed	FR3
OE-NS05	Submission	Attempt the exam again after submission	FR7
OE-NS06	Submission	Choose No on the final confirmation	FR7
OE-NS07	Cheating control	Switch tabs three times	FR9
OE-NS08	Cheating control	Right-click, copy and paste during the exam	FR9
OE-NS09	Timer	Refresh the page or press Back during the exam	FR4, FR6
OE-NS10	Exam setup	Create an exam with invalid settings	FR2
OE-NS11	Results	Open the result before it is published	FR11
OE-NS12	Access control	Candidate opens an admin page	FR12
OE-NS13	Access control	Log in on two devices at the same time	FR12

Test Cases

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
OE-NT01	OE-NS01	1. Enter a valid register number and a wrong password 2. Click Login	192312070; Wrong@1	Login fails with “Invalid register number or password”.
OE-NT02	OE-NS02	1. Leave both fields empty 2. Click Login	Blank	Mandatory-field messages are displayed.
OE-NT03	OE-NS03	1. Log in at 9:30 2. Try to click Start Exam	Time 9:30; exam starts at 10:00	Start Exam is disabled and the start time is shown.
OE-NT04	OE-NS04	1. Log in at 12:30 2. Try to start the exam	Time 12:30; window ends at 12:00	“Exam is closed” is displayed and the exam cannot be started.
OE-NT05	OE-NS05	1. Submit the exam 2. Open the same exam again 3. Click Start	Submitted exam	The second attempt is blocked with “You have already submitted this exam”.
OE-NT06	OE-NS06	1. Click Submit 2. Click No on the confirmation	Active exam	Exam stays open, and the answers and timer are unchanged.
OE-NT07	OE-NS07	1. Start the exam 2. Switch to another tab three times	3 tab switches	Warnings appear for the first two switches, and the exam is submitted automatically after the third.
OE-NT08	OE-NS08	1. Right-click on a question 2. Try to copy and paste text	Mouse and keyboard shortcuts	These actions are blocked during the exam.
OE-NT09	OE-NS09	1. Start the exam 2. Refresh the page 3. Press the browser Back button	Active exam	Timer continues from the correct remaining time, and saved answers are kept. Back does not leave the exam.
OE-NT10	OE-NS10	1. Enter duration 0 2. Set the end time before the start time 3. Add no questions 4. Save	Duration 0; end before start	Admin form shows validation errors, and the exam is not created.
OE-NT11	OE-NS11	1. Log in as a candidate after the exam 2. Open Results	Results not yet published	“Result not published” is shown and no score is displayed.
OE-NT12	OE-NS12	1. Log in as a candidate 2. Type the admin URL in the address bar	/admin/results	Access is denied, and the candidate is redirected.

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
OE-NT13	OE-NS13	1. Log in on device 1 2. Log in with the same account on device 2	Same register number	The second login is blocked, or the first session is ended, as per the policy.

Result / Conclusion

Result: 13 negative test scenarios and 13 test cases were prepared for the Online Examination System, covering 9 of the 12 functional requirements (no negative scenario for FR5, FR8, FR10). The test cases by module are:

Module	Test Scenarios	Test Cases
Login	2	2
Start exam	2	2
Submission	2	2
Cheating control	2	2
Timer	1	1
Exam setup	1	1
Results	1	1
Access control	2	2
Total	13	13

Conclusion: The negative test cases cover invalid inputs, boundary violations, failed payments or operations, and unauthorized actions in the Online Examination System. When they are executed on a build, they confirm that the system rejects wrong input safely, shows clear messages, and keeps the data and the user's session secure. Together with the positive test cases, they give a complete set of tests for the stated requirements.